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## Abstract

**Purpose:** Resistance training is one of the key components influencing power output. Previous studies directed at power development through the use of post-activation potentiation (PAP) have analyzed resistance exercises at volitional or fast tempo of movement in the entire cycle. without control of the duration of the concentric and eccentric phases of the movement. To date, no scientific studies have explored the effects of varied movement tempo on the level of power output, velocity and PAP efficiency. **Methods:** During the experimental sessions, study participants performed three sets ($Sets_{1-3}$) of the bench press (BP) exercise using 70% 1RM and two different tempos of movement: 2/0/X/0 eccentric medium tempo ($ECC_{MED}$), and 6/0/X/0 eccentric slow tempo ($ECC_{SLO}$). **Results:** Post-hoc analysis demonstrated significant differences in values of peak ($P_{PEAK}$) and mean power ($P_{MEAN}$) between $Sets_{1-3}$ measured for the $ECC_{MED}$ (2/0/X/0) tempo. The values of $P_{MEAN}$ in $Set_3$ (492.15 ± 87.61 W) were significantly higher than in $Set_2$ (480.05 ± 82.10 W) and $Set_1$ (467.65 ± 79.18 W). Similarly, the results of $P_{PEAK}$ in $Set_3$ (713.10 ± 132.72 W) were significantly higher than those obtained in $Set_2$ (702.25 ± 129.5 W) and $Set_1$ (671.55 ± 115.79 W). For the $ECC_{SLO}$ tempo (6/0/X/0) in $Set_2$ (587.9 ± 138.48 W), the results of $P_{PEAK}$ was significantly higher than in $Set_1$ (565.7 ± 117.37 W) and $Set_3$ (563.1 ± 124.93 W). **Conclusions:** The results of this study indicate that the PAP effect is observed, for both, slow and medium tempo of movement.

**Key words**: time under tension, resistance exercise, tempo, power

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## Introduction

The ability to generate power is one of the most significant factors that determines performance in numerous sport disciplines. Power output can be described by the relationship between the force generated by the muscles and movement velocity.[1] Resistance training is one of the key tools used in improving power output. Therefore, numerous studies have attempted to determine optimal variables of resistance training that would maximize power output. Particular attention in studies concerning power have been directed at exercise volume with a specific intensity of exercise.[2,3] Variations in external load and their effects on changes in power output have also been extensively explored, especially from the standpoint of long-term neuromuscular adaptations.[4,5]

The increase in power observed during and directly after strength exercises is termed post-activation potentiation (PAP). The PAP effect is evidenced as an short-term increase in power output following an activating resistance exercise. In training practice, the PAP effect may be achieved by the use of an activating exercise with a high external load (over 70%1RM), followed by an explosive activity with a similar movement structure[6]. Most reviews have concluded that moderate to heavier loads of 60-93%1RM may be most effective for producing PAP.[6,7] The physiological rationale for PAP is a greater contractibility and excitability of muscles[8] due to several mechanisms. One of these mechanisms includes increased phosphorylation of myosin light chains, rendering the actin and myosin molecules more sensitive to $Ca^{2+}$ availability, what increases power output and maximal isometric force.[9,10] Other mechanisms include higher order motoneuron recruitment due to high loading conditions, theoretically increasing fast-twitch fiber contribution, and enhanced performance in subsequent explosive activities.[11] Changes in pennation angle may also create a mechanical advantage with respect to force transmission to the tendon.[12]

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Previous studies have demonstrated a beneficial effect of initial muscle activation on power output generated in consecutive sets of strength exercises,[11,13] as well as an stimulating effect of initial muscle excitation on effectiveness of throwing, jumping and hitting movements.[6,14,15] Most studies that have analysed the acute or chronic effects of resistance training on power output have analysed training components such as load (%1RM), number of sets and repetitions, and duration of rest intervals between sets.[15,16] Among other factors that can have a significant impact on the level of power output, one must consider training variability,[15,17] training experience,[13] and muscle groups involved in the activity.[18] Previous studies concerning power output and the PAP effect postulated performance of resistance exercises at volitional or fast tempo of movement, without a precise control of the duration of both, the concentric (CON) and eccentric (ECC) phases of movement. No previous studies have established the effects of varied movement tempo, of both or either movement phases (concentric and/or eccentric cadence) on the level of power output and PAP efficiency in several consecutive sets of an resistance exercise. It has been documented that changes in movement tempo during resistance training impacts exercise volume, the level of generated force, power output, strength and hypertrophy.[19,20] Some studies have indicated that faster movement tempo in resistance training leads to increased power output,[21] whereas a slower movement tempo is more beneficial to the development of muscle strength and hypertrophy.[22,23] However, studies in this area do not relate directly to the PAP effect in various movement tempos. Furthermore, there is no scientific data, related to the acute effects of movement tempo on the PAP effect. Morrissey et al.,[24] did not demonstrate differences in power output, both in maximal and mean values between groups that trained with different tempos of the eccentric phase of movement. However, these studies considered similar eccentric duration and failed to analyse the changes in power output in several consecutive sets of an resistance exercise.

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To date it remains unclear whether there are significant differences in power output between particular sets of a resistance exercise in case of using various tempos of movement. Furthermore, previous studies have failed to establish whether the PAP effect occurs both with respect to medium and slow movement tempo. It can be assumed that the change in the tempo of movement and, consequently, the duration of work may affect the level of muscle exhaustion and, consequently, premature post-exercise fatigue, even when the number of repetitions performed is equal. The level of post-exercise fatigue affects power output in resistance exercise. Gossen and Sale[15] and Chaouachi et al.,[25] showed that the ratio of exercise volume to the duration of the rest interval is the major factor in the PAP effect, what may suggest that tempo of movement can also be an important factor determining changes in power output and PAP effect in training protocols with several consecutive sets of an activating exercise.

The PAP effect usually refers to power output, however previous studies also showed that PAP increased the 1RM load,[26] the number of repetitions performed in a set, and also allowed to increase the time under tension.[27] Thus the PAP effect observed in resistance training may be used for athletes to improve muscular strength and strength endurance performance,[28,29] therefore slow movement tempo may be effective as well. Studies analyzing the tempo of movement most often include 4-10 s time under tension values per one repetition.[20,30] A comparison of the tempo of movement with a 2 and 6-s ECC duration is justified by sports practice, as these are values often used in training by athletes.

Due to the lack of scientific data concerning the acute effects of tempo movement on power output and velocity in resistance exercises after several consecutive sets of resistance exercise, the aim of the present study was to evaluate the independent effect of medium (2/0/X/0) and slow (6/0/X/0) tempo of movement on power output ($P_{PEAK}$, $P_{MEAN}$) and bar velocity ($V_{PEAK}$, $V_{MEAN}$) in three consecutive sets of the bench press exercise (BP). The sequence of digits describing the tempo of movement (2/0/X/0 or 6/0/X/0) represents a 2 or 6-

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s eccentric (ECC) phase, 0 represents a pause during the transition phase, X represents maximum possible tempo of movement during the concentric (CON) phase, and the last digit represent no pause at the end of movement. Our initial hypothesis was that the PAP effect occurs regardless of the tempo of movement.

## Methods

All testing was performed in the Strength and Power Laboratory at the Jerzy Kukuczka Academy of Physical Education in Katowice. The experiment was performed with a randomized crossover design, where each participant performed a familiarization session with a 1RM test and two different testing protocols a week apart. During the experimental sessions, study participants performed three sets of the bench press (BP) exercise with one repetition in every set, using 70%1RM and two different tempos: 2/0/X/0 eccentric medium tempo ($ECC_{MED}$), and 6/0/X/0 eccentric slow tempo ($ECC_{SLO}$). The rest interval between sets was 5 minutes. The study participants were required to refrain from resistance training 72 hours prior to each experimental session, were familiarized with the exercise protocol and were informed about the benefits and risks of the experiment before expressing their written consent for participation in the study.

### *Subjects*

Thirty-three (33) healthy strength trained men (age = 24.0 ± 4.2 years, body mass = 77.3 ± 5.7 kg, bench press 1RM = 107,4 ± 13.5 kg; with a minimum one year of strength training experience (2.2 ± 0.57 years) took part in the research. Furthermore, the subjects participating in the experiment were expected to be able to perform a bench press with a load of at least 120% of their body mass. The participants were allowed to withdraw from the experiment at any moment, and were free of any pathologies or injuries. The study protocol was approved by the Bioethics Committee for Scientific Research, at the Academy of Physical

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Education in Katowice, Poland according to the ethical standards of the Declaration of Helsinki, 1983 (10/2018). The participants were instructed to maintain their normal dietary habits over the entire study period and not to use any dietary supplements or stimulants for the duration of the study.

*Design*

*Familiarization session and one repetition maximum test*

A familiarization session preceded the one repetition maximum testing. The 1RM test is a reliable measurement to assess muscle strength regardless of muscle group location or gender.[31] The participants arrived at the laboratory at the same time of day as the upcoming experimental sessions and cycled on an ergometer for 5 minutes at an intensity that resulted in a heart rate of approximately 130 bpm, followed by a general upper body warm-up of 10 body weight pull ups and 15 body weight push-ups. Next, the participants performed 15, 10, and 5 BP repetitions using 20%, 40%, and 60% of their estimated 1RM with a 2/0/X/0 tempo of movement. Hand placement on the barbell was set at 150% individual bi-acromial distance (BAD). The participants then executed single repetitions using tempo 2/0/X/0 with a 5 min rest interval between successful trials. The load for each subsequent attempt was increased by 2.5 kg, and the process was repeated until failure.

*Experimental sessions*

The general and specific warm-up for the experimental sessions was identical to the one used for the familiarization session. After the warm up, the participants started the main examinations and performed 3 consecutive sets of the flat bench press ($Set_1$, $Set_2$, $Set_3$) at 70%1RM and with a metronome guided tempo in the eccentric phase (Korg MA-30, Korg, Melville, New York, USA). During each set, one repetition was performed using either the 2/0/X/0/ or 6/0/X/0 tempo. The concentric phase of the BP movement was performed at

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maximal possible velocity. Power output [W] was expressed in peak values ($P_{PEAK}$) and in mean values ($P_{MEAN}$). Velocity [m/s] was measured and expressed in a similar manner in peak ($V_{PEAK}$) and mean ($V_{MEAN}$) values. A linear position transducer system "Tendo Power Analyzer" was used for the evaluation of bar velocity. The Tendo Power Analyzer is a reliable system for measuring velocity and power output.[32] The system consists of a velocity sensor connected to the barbell by a kevlar cable which, through an interface, instantly transmits the vertical velocity of the bar to a specific software installed in the computer (Tendo Software 5.0). The rest interval between sets was 5 minutes. The interval between two stages of the experiment was 7 days. Each athletes was verbally motivated and encouraged to perform with maximal intensity each of the 3 sets of the BP. All repetitions were performed without bouncing the barbell off the chest, without intentionally pausing at the transition between the eccentric and concentric phases, and without raising the lower back off the bench.

All familiarization and experimental sessions were recorded by means of a Sony camera (FDR191). The correctness of repetitions performed at a specific tempo of movement was controlled manually from the recorded data using slow speed playback.

*Statistical analysis*

The objective of the study was to evaluate differences between particular sets of the BP exercise with the $ECC_{MED}$ (2/0/X/0) and $ECC_{SLO}$ (6/0/X/0) tempo of movement. Descriptive statistics (means, standard deviations, standard errors and confidence intervals) were determined to evaluate power output [W] and velocity of the bar [m/s] expressed in mean and peak values for $ECC_{MED}$ (2/0/X/0) and $ECC_{SLO}$ (6/0/X/0), and in individual sets of the exercise ($Sets_{1-3}$). Before the analytical process, we performed the Shapiro-Wilk test where normal distribution was demonstrated at the level of significance of $p>0.05$. Significance of differences in power output (peak, mean) and bar velocity (peak, mean) between individual sets of the

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exercise ($Sets_{1-3}$) was evaluated by means of ANOVA with repeated measures and post-hoc tests. Homogeneity of variance was verified using the Levene's test and confirmed at the level of significance of $p>0.05$. Effect sizes (Cohen's d) of sets were reported where appropriate. Cohen suggested that d=0.2 be considered a 'small' effect size, 0.5 represents a 'medium' effect size and 0.8 a 'large' effect size. Statistical significance was set at $p<0.05$.

## Results

The post-hoc analysis and the effect size measurements indicate that $P_{PEAK}$ and $P_{MEAN}$ differed significantly between $Sets_{1-3}$ measured for $ECC_{MED}$ (2/0/X/0) tempo (p=0.04; d=0.83), indicating that the results of $Set_3$ were significantly higher than for $Set_1$ (p=0.04; d=0.85), whereas the results of $Set_2$ exceeded those for $Set_1$ (p=0.03; d=0.88). The results of $P_{MEAN}$ in $Set_3$ (492.15 ± 87.61 W) were significantly higher than in $Set_2$ (480.05 ± 82.10 W) and $Set_1$ (467.65 ± 79.18 W), as well $Set_2$ (480.05 ± 82.10 W) were significantly higher than $Set_1$ (467.65 ± 79.18 W). The results of $P_{PEAK}$ in $Set_3$ (713.10 ± 132.72 W) were significantly higher than those obtained in $Set_2$ (702.25 ± 129.5 W) and $Set_1$ (671.55 ± 115.79 W), as well as $Set_2$ (702.25 ± 129.5 W) were significantly higher than $Set_1$ (671.55 ± 115.79 W) (tab.1).

Furthermore, with the $ECC_{SLO}$ tempo (6/0/X/0) results of $P_{PEAK}$ in $Set_2$ (587.9 ± 138.48 W) were significantly higher (p=0.04; d=0.85) than those obtained in $Set_1$ (565.7 ± 117.37 W), and $Set_3$ (563.1 ± 124.93 W). No significant differences were found between $Sets_{1-3}$ in $P_{MEAN}$ for the $ECC_{SLO}$ (6/0/X/0) tempo ($p>0.05$), (tab.2).

The levels of $V_{PEAK}$, $V_{MEAN}$ did not differ significantly between $Sets_{1-3,}$ both for $ECC_{MED}$ (2/0/X/0) and $ECC_{SLO}$ tempo of movement (6/0/X/0) ($p>0.05$) (tab.1, 2).

## Discussion

The major finding of the present study is that there were significant differences in power output between particular $Sets_{1-3}$ in tempo 2/0/X/0, both with regards to $P_{PEAK}$ and $P_{MEAN}$. In

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tempo 6/0/X/0 there were also significant differences but only in regards to $P_{PEAK}$. During the 6/0/X/0 tempo $P_{PEAK}$ in $Set_2$ was significantly higher compared to $Set_1$, what indicates that the PAP effect also occurs during the slow movement tempo in the ECC phase of movement. The study showed no significant differences in $V_{PEAK}$ and $V_{MEAN}$ between particular sets for both fast $ECC_{MED}$ and slow $ECC_{SLO}$ cadences.

These results are consistent with findings presented by Morales-Artacho et al.,[33] who documented increases in power output following resistance training in the second and third sets compared to the first one. The increase in power output ($P_{PEAK,}$ $P_{MEAN}$) in $Set_2$ and $Set_3$ compared to $Set_1$ for the $ECC_{MED}$ tempo, and the increase in power output ($P_{PEAK}$) in $Set_2$ for $ECC_{SLO}$ tempo suggests the occurrence of the PAP effect. The post-activation increase in power output in consecutive BP sets is likely to have an effect on sensitivity of actin and myosin to $Ca^{2+}$ ions availability following the activation stimuli in which the first set was performed.[9,10,14]. The increase in power output may also be attributable to the elevated recruitment of higher order motor neurons that increase contribution of fast-twitch fibres in subsequent sets of a specific movement tempo.[34] The effects of PAP observed in our study is consistent with findings of previous studies that demonstrated an increase in muscle strength and power output following resistance exercise with moderate and high external loads.[12,14,34] Our study shows that both, medium (2/0/X/0) and slow (6/0/X/0) tempo of movement lead to an increase in power output in consecutive sets. In case of $ECC_{MED}$ tempo, we demonstrated higher values of power output in both, $Set_2$ and $Set_3$ compared to $Set_1$ ($P_{MEAN,}$ $P_{PEAK}$). However, in case of the $ECC_{SLO}$ tempo, only the value of power output ($P_{PEAK}$) for $Set_2$ was significantly higher than for $Set_1$. The level of peak power output generated in $Set_3$ for $ECC_{SLO}$ tempo was significantly lower than in $Set_2$. A slower ECC phase of movement reduces power output[35] and diminishes the PAP effect during resistance exercise and is not preferable for power training. Furthermore, the stretch-shortening cycle (SSC) is another mechanisms that may explain the obtained results.

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A slower ECC phase can inhibit the SSC effect,[36] yet it is difficult to judge on this basis whether the higher values of power output in subsequent sets during $ECC_{MED}$ tempo result from the PAP effect or from a more effective use of the SSC. However the occurrence of the PAP effect after a slow movement tempo may be suitable and effective for improving maximal strength and strength endurance performance.[27,28,29]

The decline in power output in $Set_3$ for the $ECC_{SLO}$ may be linked to the time under tension (TUT) in particular repetitions. The TUT in the $ECC_{SLO}$ tempo was three times higher compared to $ECC_{MED}$. TUT represents the time of effort, which by some researchers is referred to as the volume of resistance training.[20] According to the results of Sakamoto and Sinclair[26] the longer the repetition duration, the lower is the number of performed repetitions for the same external load. The three times higher TUT for the $ECC_{SLO}$ tempo can lead to greater muscle exhaustion and, consequently, premature post-exercise fatigue. It is likely that the fatigue following $Set_1$ and $Set_2$ with the $ECC_{SLO}$ tempo led to a decline in the power output in $Set_3$. However in this study, significant changes were observed with respect to power output rather than velocity. The evaluations did not demonstrate significant changes in $V_{PEAK}$ and $V_{MEAN}$ both in the $ECC_{MED}$ and $ECC_{SLO}$ tempo. However, it is worth noticing that with the $ECC_{MED}$ tempo, higher values of $V_{PEAK}$ were reached in $Set_2$ (0.70 ± 0.15m/s) compared to $Set_1$ (0.69 ± 0.13m/s) and $Set_3$ (0.68 ± 0.13m/s). Similar results were observed for the $ECC_{SLO}$ tempo, and it was noticeable that the results of $V_{PEAK}$ in $Set_2$ were higher than those for $Set_1$ yet in $Set_3$ they were lower than those for $Set_2$. Because power output is the product of force and velocity, both power output and velocity need to be controlled during a resistance training program.

In addition to exercise volume or time of effort, duration of rest periods between the sets also have a significant effect on maintaining the level of power output. An optimal rest interval should ensure the balance between fatigue and performance.[15] Due to a three-times higher level of TUT in the $ECC_{SLO}$ compared to $ECC_{MED}$ tempo, the rest duration between

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particular sets could have been insufficient, which affected the power output in $Set_3$ during the $ECC_{SLO}$ tempo. On the contrary, with the $ECC_{MED}$ tempo, a significant increase in power output, both mean and peak, was found in $Set_3$ compared to $Set_1$ and $Set_2$. Previous studies have found that optimal effects of PAP in the BP occurred after 7 minutes of rest following the activating exercise at 1RM.[34] Because in our research a lower value of 1RM was used compared to the study by Ferreira et al.,[34] we decided to use a shorter rest interval (5 min). In case of the $ECC_{MED}$ tempo, the rest interval was sufficient to reach equilibrium between fatigue and potentiation for each of the 3 consecutive sets. However, with the $ECC_{SLO}$ tempo, the rest interval was insufficient with respect to the work performed. Despite similar total work (70%1RM x 1rep x 3set) for both $ECC_{SLO}$ and $ECC_{MED}$ tempo, the total time under tension was three time longer for the $ECC_{SLO}$ compared to $ECC_{MED}$ tempo which indicates that TUT can be a variable defining the volume of work regardless of the number of performed repetitions.[20,37]. The results are consistent with indications of Gossen and Sale[15] and Chaouachi et al.,[25] who showed that the ratio of exercise volume to rest duration is the major factor in post-activation potentiation.

The experiment demonstrated that the ratio of exercise volume to rest duration is important for the optimal development of PAP for both $ECC_{MED}$ and $ECC_{SLO}$ tempo. In addition to exercise volume, Morales-Artacho et al.,[33] also found that a high external load may limit the PAP effect in several consecutive sets of a resistance exercise. However, the results of our study are inconsistent with those of Morales-Artacho et al.,[33] in which an increase in power output in consecutive sets was obtained despite using a high external load (70%1RM). Another factor that impacts the balance between the rest interval, the amount of performed work and the power output generated during the exercise is the training experience of study participants.[11,13] The three-time higher TUT for $ECC_{SLO}$ tempo, insufficient rest durations between sets, and greater training experience most likely led to the decline in power output in

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$Set_3$ with the $ECC_{SLO}$ tempo compared to the $ECC_{MED}$ tempo in which an increase in power output was observed.

The limitation of the present study pertains to the one repetition test. The 1RM was performed using a 2/0/X/0 tempo, and an independent 1RM test for the 6/0/X/0 tempo was not performed. Headley et al.,[19] showed significant differences in the maximal load (1RM) using a different tempo of movement. However the aim of this study was to analyse changes during successive sets without determination of changes between tempos, therefore, any differences in 1RM between the 2/0/X/0 and 6/0/X/0 tempo should not have affected the obtained results.

**Practical implications**

The post-activation potentiation effect occurs for each ($ECC_{MED}$ and $ECC_{SLO}$) movement tempo, whereas the changes in power output in particular sets depend on the ratio of exercise volume or time of the effort to the duration of the rest interval between sets, regardless of movement tempo. The results of this study demonstrated the validity of accumulation of sets in the training unit and necessity of a direct control of each set in striving for optimal development of power output. Furthermore the study indicates that training focused on power development with the use of the PAP effect, should concentrate on maximal tempo of the CON phase of movement, yet a fast tempo in the ECC phase of movement is also preferred. This is especially important when using higher external loads (>70%1RM), where inhibition (breaking) in the ECC movement has been often observed in practice, regardless of the athlete's safety and elevated injury risks.

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## Conclusions

This study demonstrates that the PAP effect is observed for both medium (2/0/X/0) and slow (6/0X/0) tempo of movement. The slow tempo of movement may be preferable for the development of maximal strength and strength endurance.

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**Table 1:** Levels of power output and velocity during tempo 2/0/X/0.

| | | Set 1 | Set 2 | Set 3 |
|---|---|---|---|---|
| Power [W] | Mean | 467.65±79.18*# | 480.05±82.10*† | 492.15±87.61†# |
| Power [W] | Peak | 671.55±115.79*# | 702.25±129.50*† | 713.10±132.72†# |
| Velocity [m/s] | Mean | 0.60±0.09 | 0.62±0.09 | 0.62±0.10# |
| Velocity [m/s] | Peak | 0.79±0.10 | 0.82±0.10 | 0.81±0.11 |

*#† statistically significant differences

**Table 2:** Levels of power output and velocity during tempo 6/0/X/0.

| | | Set 1 | Set 2 | Set 3 |
|---|---|---|---|---|
| Power [W] | Mean | 401.95±65.42 | 400.05±73.05 | 397.00±83.20 |
| Power [W] | Peak | 565.70±117.37* | 587.90±138.48*# | 563.10±124.93# |
| Velocity [m/s] | Mean | 0.52±0.08 | 0.52±0.09 | 0.51±0.09 |
| Velocity [m/s] | Peak | 0.69±0.13 | 0.70±0.15 | 0.68±0.13 |

*# statistically significant differences

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